Fresh explicit-one-worker guard confirmation

Run 20261019-040338-boolean-native-serial-gu-6ec5e9. Fresh same-binary confirmation after the initial four-worker regression. Five baseline A/A and five serial0/1 pairs per cell; each one-worker round additionally measures the legacy kernel with all three switches off. Exact input, raw/canonical output, rank, XOR and profile counts match. No host or earlier sample is pooled.

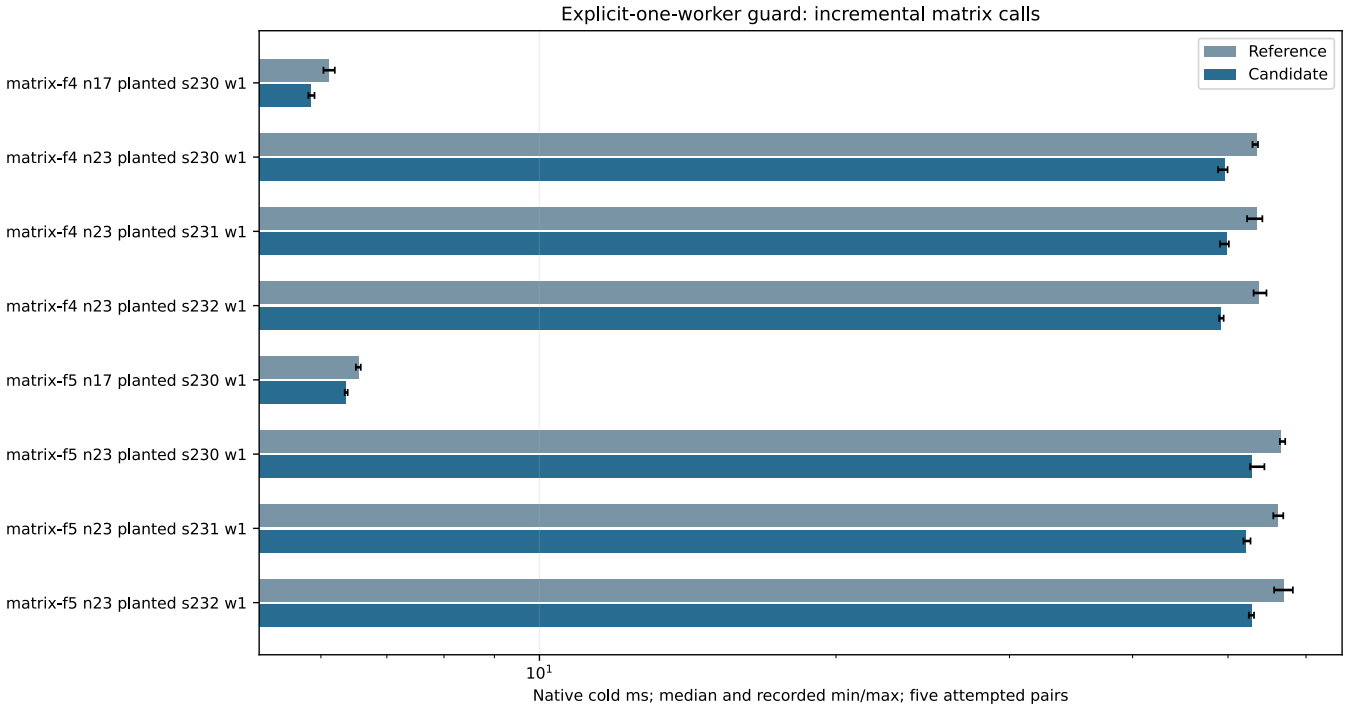
Host: Intel(R) Xeon(R) Platinum 8488C; rustc 1.99.1 (58cf8f9c3 2026-10-13); c7i.2xlarge on-demand; reserved cores/SMT siblings: one=3,7 four=3,7,2,6. The refined guard caches the explicit RAYON_NUM_THREADS=1 configuration and checks the actual current pool before selecting serial work. With an explicit four-worker configuration, both switch states skip the registry query.

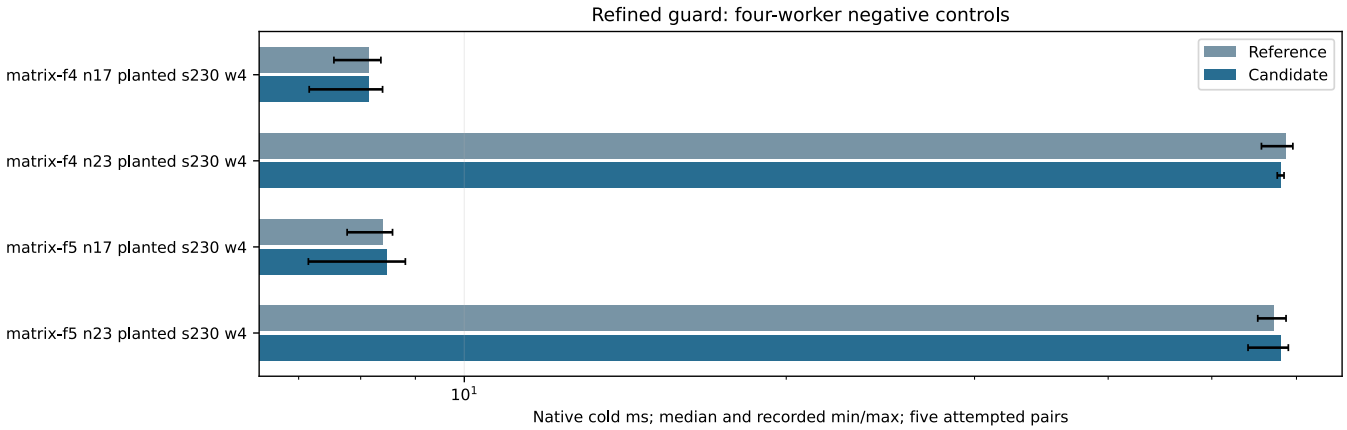
All four multiworker controls are neutral under the frozen gate.

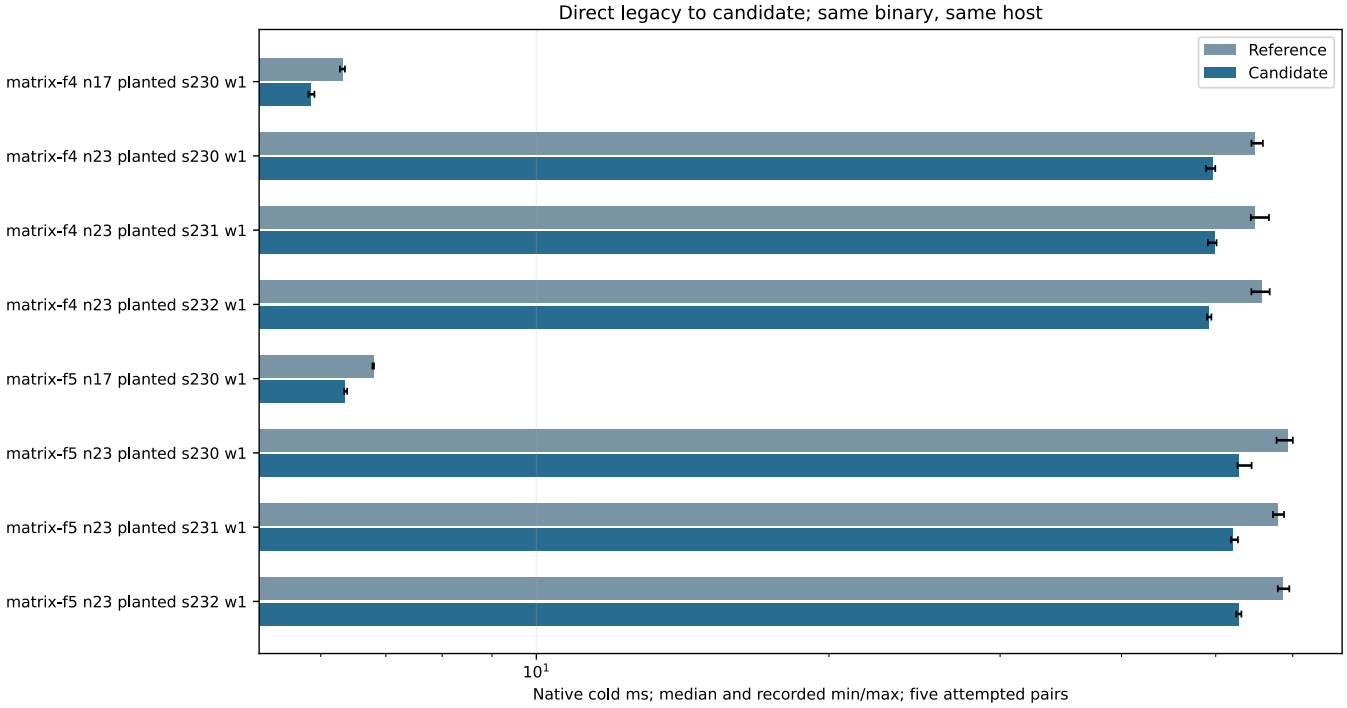
All eight one-worker incremental cells qualify faster. Primary n23 ratios are1.0802 F4 and1.0730 F5. Direct same-host legacy to candidate ratios are1.1141 and1.1209, with CIs[1.0980,1.1297] and[1.0608,1.1402]. All12cells have five eligible pairs; eleven rejected attempts remain archived. These ratios do not reprice the earlier complete-solver table.

Incremental pairs fix single-write/vector switches1 in both arms. Direct legacy ratios compare that round's all-off kernel with the candidate in the same binary, rather than multiplying previous gains. Admission, first eligible of three attempts, exact3125-resample CI and max(2%, twice A/A noise) are unchanged. Every slower, neutral or missing outcome remains in serial-guard-summary.json.

The switch remains opt-in and requires explicit one-worker configuration; nested multiworker pools keep parallel iteration. Four worker/switch configurations pass the existing GF2 reference tests before native timing. These are matrix calls, not complete recursive roots or decomposition. The eighteen-arm cost table retains its earlier source and serial-clear-off flags. Source: SERIAL_GUARD_PROTOCOL.md, run-serial-guard.sh, analyze-serial-guard.rs and runs/serial-guard/ in the raw archive. The prior slower control is preserved in its own report.




Refined guard: four-worker negative controls
Reference
Candidate
matrix-f4 n17 planted s230 w4
matrix-f4 n23 planted s230 w4
matrix-f5 n17 planted s230 w4
matrix-f5 n23 planted s230 w4
10^1
Native cold ms; median and recorded min/max; five attempted pairs


Direct legacy to candidate; same binary, same host
Reference
Candidate
matrix-f4 n17 planted s230 w1
matrix-f4 n23 planted s230 w1
matrix-f4 n23 planted s231 w1
matrix-f4 n23 planted s232 w1
matrix-f5 n17 planted s230 w1
matrix-f5 n23 planted s230 w1
matrix-f5 n23 planted s231 w1
matrix-f5 n23 planted s232 w1
10^1
Native cold ms; median and recorded min/max; five attempted pairs

Incremental pairs and four-worker controls

Arm	n	Seed	Workers	Before ms	After ms	Ratio	Decision
matrix-f4	17	230	1	6.1083	5.8542	1.0434	faster
matrix-f4	23	230	1	53.4171	49.6467	1.0802	faster
matrix-f4	23	231	1	53.4533	49.8884	1.0695	faster
matrix-f4	23	232	1	53.7325	49.1784	1.0913	faster
matrix-f5	17	230	1	6.5614	6.3594	1.0276	faster
matrix-f5	23	230	1	56.6003	52.8025	1.0730	faster
matrix-f5	23	231	1	56.1721	52.0773	1.0721	faster
matrix-f5	23	232	1	56.9634	52.8105	1.0853	faster
matrix-f4	17	230	4	8.1520	8.1383	1.0093	neutral
matrix-f4	23	230	4	58.7065	58.0531	1.0180	neutral
matrix-f5	17	230	4	8.3863	8.4727	0.9863	neutral
matrix-f5	23	230	4	57.1523	58.0442	0.9974	neutral

Direct legacy to candidate; one worker

Arm	n	Seed	Legacy ms	Candidate ms	Ratio	Decision
matrix-f4	17	230	6.3206	5.8542	1.0813	faster
matrix-f4	23	230	54.9114	49.6467	1.1141	faster
matrix-f4	23	231	54.8660	49.8884	1.1005	faster
matrix-f4	23	232	55.7328	49.1784	1.1320	faster
matrix-f5	17	230	6.8030	6.3594	1.0691	faster
matrix-f5	23	230	59.2979	52.8025	1.1209	faster
matrix-f5	23	231	57.9081	52.0773	1.1170	faster
matrix-f5	23	232	58.6563	52.8105	1.1057	faster